



भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

CS6903: Network Security (Introduction to Scapy)

Saurabh Kumar
CSE Department, IIT Hyderabad

Acknowledgement

- ❑ Scapy Documentation
- ❑ Scapy Python-Based Interactive Packet Manipulation, presentation at BlackHat USA 2019 by Guillaume Valadon (https://guedou.github.io/talks/2019_BHUSA/Scapy.slides.html#/)
- ❑ Various other sources on Internet

What is Scapy?

- ❑ Scapy is a packet manipulation tool for computer networks.
 - forge or decode packets, send them on the wire, capture them, and match requests and replies
- ❑ Allows construction of tools that can handle task such as
 - Scanning
 - tracerouting
 - attack
 - network discovery
 - and many more
- ❑ Requires **super user (root)** privileges

Scapy: Packets as Python Objects

- ❑ Each layer packet is termed as Python Object
- ❑ Packet Corresponding to Each Layer
 - MAC: Ether()
 - Network: IP()
 - Transport: TCP() or UDP()
 - Application: HTTP() and many more
- ❑ / operator is used to stack the layers

Supported Protocols

- ❑ ls to check supported protocols

```
>>> ls()
```

- ❑ Details about a specific protocol

```
>>> ls(IP)
```

- ❑ Commands/functions available

```
>>> lsc()
```

Crafting Packets

❑ SYN/ACK Packet (example IP: 192.168.0.1)

```
>>> pkt = IP(dst="192.168.0.1")
```

```
>>> pkt /= TCP(dport=80, flags="SA")
```

❑ ICMP Host Unreachable packet

```
>>> pkt = IP(dst="192.168.0.1")
```

```
>>> pkt /= ICMP(type=3, code=1)
```

Sending and Receiving Packets

□ send (or sendp): Send packet at L3 (or L2)

>>> send(packets)

□ sr (or srp): send packets at L3 (or L2) and receive all responses

>>> ans, unans = sr(packets)

□ sr1 (or srp1): send packets at L3 (or L2) and receive one response

>>> resp = sr1(packets)

Displaying Packets

- Summary of each packets

```
>>> packets.summary()
```

- Whole packet list

```
>>> packets.show()
```


Port Scanning: TCP SYN

- ❑ Only single port

```
>>> resp = sr1(IP(dst="192.168.0.1") / TCP(dport=80, flags="S"))
```

- ❑ Scanning multiple ports

```
>>> ans, unans = sr(IP(dst="192.168.0.1") / TCP(dport=(80,100),  
flags="S"))
```

Packet Sniffing

❑ Configure the network interface to sniff packets from

```
>>> conf.iface = "eth0"
```

❑ Configure sniffer to sniff packets

```
>>> packets = sniff(count=20) # count=0 means infinite packets
```

Read and Write Packets from/to .pcap File

- Read .(p)cap file

```
>>> packets = rdpcap("path_to_pcap_file")
```

- Write packets to .(p)cap file

```
>>> wrpcap("file_name", packets)
```

Task To Do

❑ Create

- TCP ACK Port Scanner
- TCP Replay Tool
- UDP Ping Tool
- Sniffer that filters based on user input
- tracer
- arpping
- Arppoisoning
- Network Mapper Script

❑ ***Don't use Scapy inbuilt tools.**